KAIZEN MATHS

Lesson starter

Logarithms and Exponential Equations

Logarithm Laws and Operations

Name _____

S1

Simplify: $\log_5(5) + \log_5(4)$

S2

Simplify: $\log_2(12) - \log_2(3)$

S3

Simplify: $\log_3(5^2)$

S4

Evaluate: $\ln(e^2)$

Show your working

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Exit ticket

Logarithms and Exponential Equations

Logarithm Laws and Operations

Name _____

E1

Simplify: $\log_5(9) + \log_5(10)$

E2

Simplify: $\log_{10}(20) - \log_{10}(5)$

E3

Simplify: $\log_5(5^4)$

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Teacher answers & working

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S1

Simplify: $\log_5(5) + \log_5(4)$

Answer:

$\log_5(20)$

Working:

Method focus: Combine logarithms only when the bases match and the arguments are valid.

Identify the rule

Adding logs with the same base uses the product rule

Apply the product rule

$\log_a(x) + \log_a(y) = \log_a(xy)$

Multiply the arguments

$5 \times 4 = 20$

$\log_5(20)$

Final answer: $\log_5(20)$

S2

Simplify: $\log_2(12) - \log_2(3)$

Answer:

$\log_2(4)$

Working:

Method focus: Combine logarithms only when the bases match and the arguments are valid.

Identify the rule

Subtracting logs with the same base uses the quotient rule

Apply the quotient rule

$\log_a(x) - \log_a(y) = \log_a(x/y)$

Divide the arguments

$12 / 3 = 4$

$\log_2(4)$

Final answer: $\log_2(4)$

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S3

Simplify: $\log_3(5^2)$

Answer:

$2\log_3(5)$

Working:

Method focus: Combine logarithms only when the bases match and the arguments are valid.

Identify the exponent inside the logarithm

The argument is 5^2

Apply the power rule

$\log_a(x^n) = n\log_a(x)$

Bring the exponent down

$\log_3(5^2) = 2\log_3(5)$

$2\log_3(5)$

Final answer: $2\log_3(5)$

S4

Evaluate: $\ln(e^2)$

Answer:

2

Working:

Method focus: Natural logarithms use base e.

Recall the meaning of natural log

$\ln(x) = \log_e(x)$

Rewrite the argument as a power of e

$e^2 = e^2$

Apply $\log_e(e^n) = n$

$\ln(e^2) = 2$

2

Final answer: 2

Teacher answers & working

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E1

Simplify: $\log_5(9) + \log_5(10)$

Answer:

$\log_5(90)$

Working:

Method focus: Combine logarithms only when the bases match and the arguments are valid.

Identify the rule

Adding logs with the same base uses the product rule

Apply the product rule

$\log_a(x) + \log_a(y) = \log_a(xy)$

Multiply the arguments

$9 \times 10 = 90$

$\log_5(90)$

Final answer: $\log_5(90)$

E2

Simplify: $\log_{10}(20) - \log_{10}(5)$

Answer:

$\log_{10}(4)$

Working:

Method focus: Combine logarithms only when the bases match and the arguments are valid.

Identify the rule

Subtracting logs with the same base uses the quotient rule

Apply the quotient rule

$\log_a(x) - \log_a(y) = \log_a(x/y)$

Divide the arguments

$20 / 5 = 4$

$\log_{10}(4)$

Final answer: $\log_{10}(4)$

Teacher answers & working

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E3

Simplify: $\log_5(5^4)$

Answer:

$4\log_5(5)$

Working:

Method focus: Combine logarithms only when the bases match and the arguments are valid.

Identify the exponent inside the logarithm

The argument is 5^4

Apply the power rule

$\log_a(x^n) = n\log_a(x)$

Bring the exponent down

$\log_5(5^4) = 4\log_5(5)$

$4\log_5(5)$

Final answer: $4\log_5(5)$